Syrian Arab Republic

Ministry of Higher Education

Syrian Virtual University

Bachelor of IT Program BAIT

MS SQL Server Administration IIS404

Chapter three

High Availability & Disaster Recovery

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Contents

1. Intro

Armed with the knowledge that everything can fail; you should build in redundancy where possible. The sad reality is that these decisions are governed by budget constraints. The amount of money available is inversely proportional to the amount of acceptable data loss and length of downtime. For business-critical systems, however, uptime is paramount, and a highly available solution will be more cost effective than being down, considering the cost-per-minute to the organization.

Availability can be measured relative to "100% operational". A widely-held but difficult-to-achieve standard of availability for a system or product is known as "five 9s" (99.999 percent) availability.

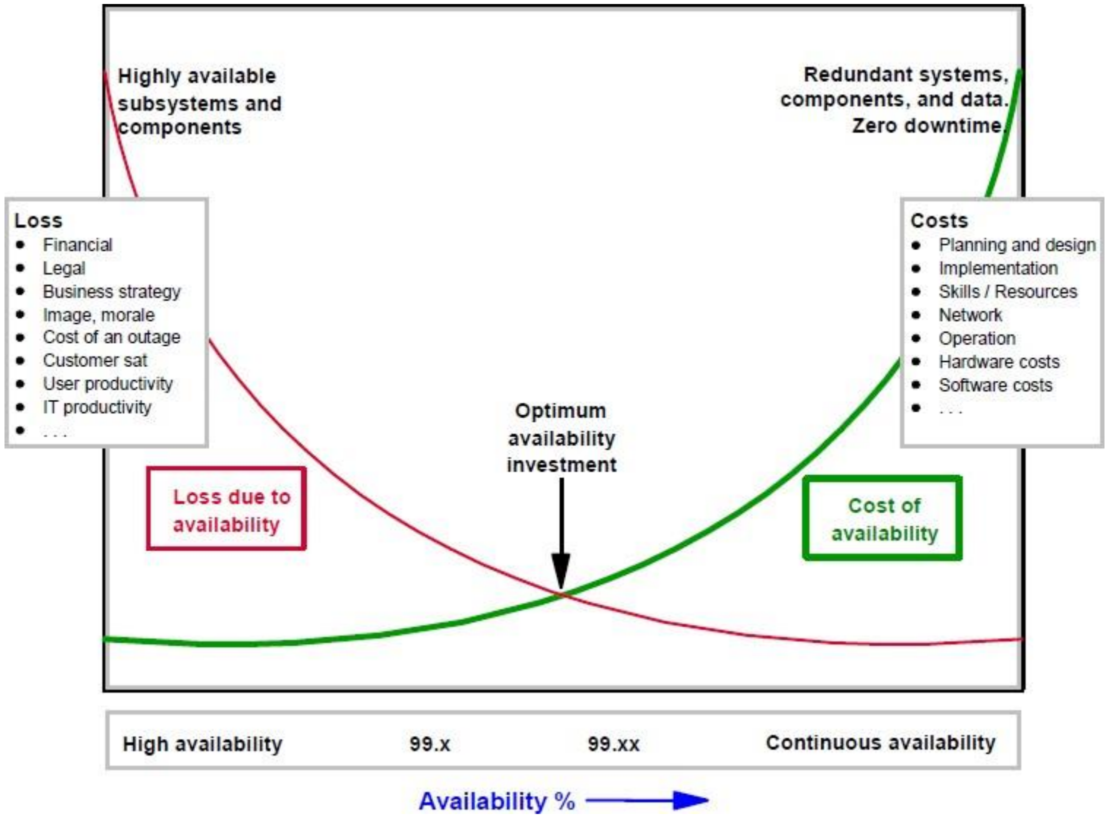
It is nearly impossible to guarantee zero downtime with zero data loss. There is always a trade-off. The business decides on that trade-off, based on resources (equipment, people, money), and the technical solution is in turn developed around that trade-off. The business drives this strategy using two values called the Recovery Point Objective and Recovery Time Objective, which are defined in a Service-Level Agreement (SLA).

Recovery Point Objective

A good way to think of Recovery Point Objective (RPO) is “How much data are you prepared to lose?” When a failure occurs, how much data will be lost between the last transaction log backup and the failure? This value is usually measured in seconds or minutes.

Recovery Time Objective

The Recovery Time Objective (RTO) is defined as how much time is available to bring the environment up to a known and usable state after a failure. There might be different values for HA and disaster recovery scenarios. This value is usually measured in hours.



2. High Availability (HA) vs. Disaster Recovery (DR)

يجب الأخذ بعين الإعتبار معياري ال RTO, (الزمن السموح إستغراقه لعودة الخدمة للعمل) RPO (كمية البيانات السموح ضياعها

- **) عند العمل على ال HA, DR**
- **كل ما خفّضنا ال Down Time سيتطلب ذلك تكلفة مادية أعلى**
- **فكرة ال HA فكرة وقائية تعتمد على إبقاء السيرفر Online قدر الإمكان (Available)**

بينما فكرة ال DR فكرة علاجية

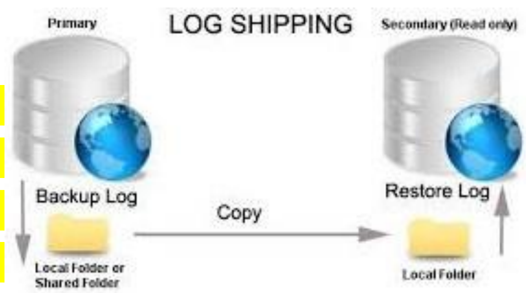
HA is not disaster recovery (DR). They are often grouped under the same heading (HA/DR), mainly because there are shared technology solutions for both concepts, but HA is about keeping the service running, whereas DR is what happens when the infrastructure fails entirely. DR is like insurance: you don’t think you need it until it’s too late. HA costs more money, the shorter the RPO.

- HA goal is to mitigate the downtime of the system.
- DR deals with situations when a system needs to be recovered after a disaster, which was not prevented by the high availability strategy in use.

3. SQL Server HA & DR Solutions

- Log shipping (DR almost)
- Replication (DR & HA partially)
- Database mirroring (will be removed in future versions, DR)
- Always On Failover Cluster Instances
- Always On availability groups

3.1. Log Shipping



تعتمد على فكرة أن ال Server الأول (ال Primary) نريد أن نعمل منه نسخ متاحة للإستخدام في حالة وقوعه و توقفه عن العمل , حيث نعمل Replication ل Log files عبر نسخها إلى Shared Folder ممكن للسيرفر الثاني أو الثالث القيام ب Restore Log منه

حيث عند توقّف ال Primary Server عن العمل نقوم بالإنتقال للثاني أو الثالث بشكل يدوي

- يقوم ال SQL Agent بجدولة أعمال العمليات (Jobs) بين ال Servers:
  1) Backup لل Log من ال Server الأول لمجلد مشترك
  2) نقل ال Backup من المجلد المشترك لمجلد المتشرك لل Server الثاني
  3) Restore من المجلد المشترك للسيرفر الثاني لل Server

و نضع بعدها ال Secondary Server إحدى الحالتين (Standby يعني لا يعمل و لكن جاهز للعمل في حال وقوع السيرفر الأساسي – أو Read-Only ممكن إستخدامه لل Backups أو أي خدمة ثانية ....)

- تكون طريقة ال Log Shipping متاحة على كل نسخ ال SQL Server و متاحة على اللينكس
- يجب أن نتذكر أن نوع ال Simple Recovery لن يعمل معنا في هذه الحالة, لانه لا يقوم ب Recover لل Log
- يطلق على عملية الإنتقال للسيرفر الثاني أو الثالث في حال وقوع الأول ب Failover و تكون يدوية
- مثال عن ال Log Shipping:

```
Use master
Go

Drop database if exists shipDB
Go

Create database shipDB on
( name='ShipDB_Data' , Filename='E\svu\bait\..\ShipDB_data.mdf' )
Log on    (name='ShipDB_Log' , Filename=' E\svu\bait\....ShipDB_log.ldf' )

Go
```

Use ShipDB
Go

Create table shiptable (id int primary key)

نقوم ب Backup على السيرفر الأول إلى ال Shared Folder:

Backup Database ShipDB to Disk = ‘E:\svu\bait\Demo|…\ShipDBFB.bak ‘

ثم على السيرفر الثاني, نقوم ب Restore لل Database من ال Shared Folder و نحدد أن تكون عملية ال Restore من نوع : With Norecovery

Restore Database ShipDB from Disk = ‘E:\svu\bait\Demo|…\ShipDBFB.bak ‘ with norecovery

ثم نقوم بنقل ملفات ال Database إلى مكان خاص بالسيرفر الثاني:

Move “ShipDB_Data’ to ‘G:\new_DB\...\ShipDB_mirror_data.mdf’,
Move “ShipDB_Data’ to ‘G:\new_DB\...\ShipDB_mirror_log.ldf’,
Norecovery

و نستطيع القيام بالعمليات السابقة من Backup و Restore من و إلى ال Shared Folder بإستخدام ال GUI كما قام الدكتور في المحاضرة أيضا

و يجب ملاحظة أنه عند الوصول لخطوة ال Restore لل DB عن طريق ال GUI يتم سؤالنا عن الوضع المراد وضعها به بعد عملية ال Restore, حيث يكون لدينا وضعين متاحين: 1) No Recovery mode: يعني إبقائها Offline 2) Standby mode: يعني وضعها بحالة Read Only

نقوم بإدخال بعض المعلومات لل DB في السيرفر الأول:

Insert into shiptable values (1000)

ثم ننتظر 15 دقيقة ليقوم السيرفر الأول بعملية Backup مأتمتة

ال Job في ال SQL Server Agent هو المسؤول عن أتمتة العمليات و جدولتها, مثل جدولة عملية ال Backup حيث يكون الوقت الإفتراضي لل Backup كل 15 دقيقة, و لإجراء Backup فوري نقوم بالدخول إلى Jobs من منسدلة ال SQL

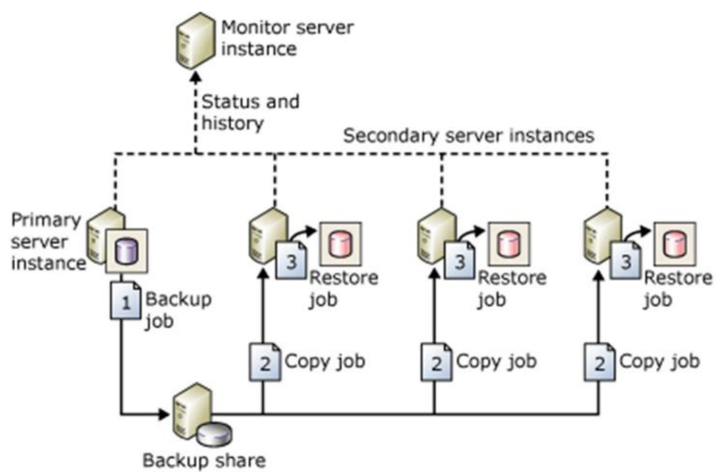
Server Agent و من ثم الضغط باليمين على ال LSBackup_ShipDB و إختيار Backup فوري

في السيرفر الثاني, بعد إنتظار 15 دقيقة للقيام بعملية Restore مؤتمتة, أو القيام بها يدويا بتفعيل ال Job المسؤول عنها يدويا ثم نتحقق من وجود نفس المعلومات المدخلة في السيرفر الأول لدينا:

Use ShipDB
Select * from shiptable

Like Always On availability groups and database mirroring, log shipping operates at the database level. SQL Server Transaction Log Shipping is an extremely flexible technology to provide a relatively inexpensive and easily managed HA and DR solution.

The principle is as follows: a primary database is in either the Full or Bulk Logged recovery model, with transaction log backups being taken regularly every few minutes. These transaction log backup files are transferred to a shared network location, where one or more secondary servers restore the transaction log backups to a standby database.



A log shipping configuration does not automatically fail over from the primary server to the secondary server. If the primary database becomes unavailable, any of the secondary databases can be brought online manually. You can use a secondary database for reporting purposes. In addition, you can configure alerts for your log shipping configuration.

If you use the built-in Log Shipping Wizard in SQL Server Management Studio, on the Restore tab, click Database State When Restoring Backups, and then choose the No Recovery Mode or Standby Mode option (*https://docs.microsoft.com/sql/database-engine/logshipping/configure-log-shipping-sql-server*).

If you are building your own log shipping solution, remember to use the RESTORE feature with NORECOVERY, or RESTORE with STANDBY.

If a failover occurs, the tail of the log on the primary server is backed up the same way (if available—this guarantees zero data loss of committed transactions), transferred to the shared location, and restored after the latest regular transaction logs. The database is then put into RECOVERY mode (which is where crash recovery takes place, rolling back incomplete transactions and rolling forward complete transactions).

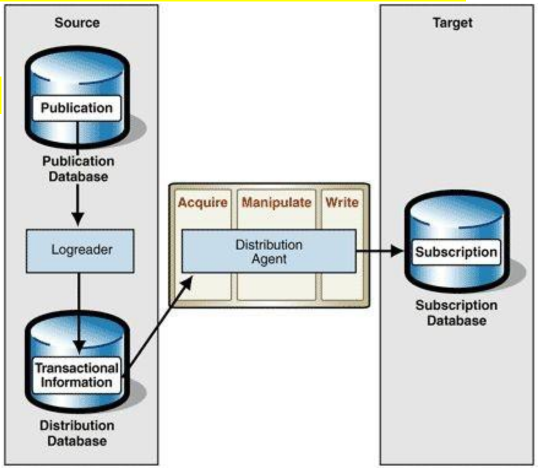
As soon as the application is pointed to the new server, the environment is back up again with zero data loss (tail of the log was copied across) or minimal data loss (only the latest shipped transaction log was restored).

Log Shipping is a feature that works on all editions of SQL Server, on Windows and Linux. However, because Express edition does not include the SQL Server Agent, Express can be only a witness, and you would need to manage the process through a separate scheduling mechanism. You can even create your own solution for any edition of SQL Server, using Azure Blob Storage and AzCopy.exe, for instance.

3.2. Replication

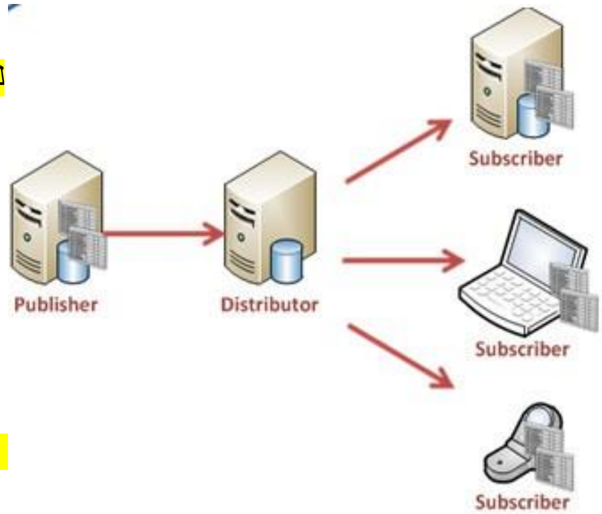
تعتمد على مبدأ وجود Master Database واحدة موجودة عند ال Publisher

يتم بها إنشاء ال ***Customers*** و نقلها إلى الافرع الأخرى (ال ***Subscribers***) عن طريق ال ***Distributor***



Replication is a set of technologies for copying and distributing data and database objects from one database to another and then synchronizing between databases to maintain consistency.

Use replication to distribute data to different locations and to remote or mobile users over local and wide area networks, dial-up connections, wireless connections, and the Internet.



له ثلاث أنوع:

1) ***Snapshot Replication***:
يتم نقل كل ال ***Data*** من ال ***Distributor*** إلى ال ***Subscriber*** (تتم عملية ***Overwrite***)
2) ***Transactional Replication***
3) ***Merge Replication***

3.2.1. Snapshot Replication

خصائصها:

1) **تكون المزامنة بإتجاه واحد من ال Publisher و ال Distributor بإتجاه ال Subscribers**
2) **تكون كل فترة معيّنة**
3) **لا تكون المعلومات Up-to-date عند ال Subscribers**

- يكون ال **Snapshot Replication** الخيار الأول عند إستخدام أي **Replication**
- ممكن تعيين ال **Subscribers** ك **Publishers** في حالات معيّنة لكي يقومو بإرسال جداول معيّنة مع معلوماتهم إلى ال **Publisher** الذي يعمل ك **Subscriber** في هذه الحالة
-

Provides the initial data set for transactional and merge replication; It can also be used when complete refreshes of data are appropriate.

Using snapshot replication by itself is most appropriate when one or more of the following is true:

- Data changes infrequently.
- It is acceptable to have copies of data that are out of date with respect to the Publisher for a period of time.
- Replicating small volumes of data.
- A large volume of changes occurs over a short period of time.

3.2.2. Transactional Replication

Applied where high latency is not allowed, such as an OLTP system for a bank, because you always need real-time updates of cash or stocks.

Typically used in server-to-server scenarios that require high throughput, including:

- improving scalability and availability;
- data warehousing and reporting;
- integrating data from multiple sites;
- integrating heterogeneous data;

3.2.3. Merge Replication

- تكون بإتجاهين
- يقبل عدم وجود **PK** لكل جدول, حيث سيتم إنشاء **GUID** لكل جدول و سيتم الإعتماد عليه
- تتميز بأن ال **Publisher** و ال **Subscriber** سيتبقى تعمل ك **Publisher** و **Subscriber** طول الوقت مع إستخدام **Replication** بالإتجاهين

مثال عن ال **Merge Replication**:

```
Use master
Go

Drop database if exists IIS404R
Go

Create database IIS404R on
( name='IIS404R_Data' , Filename='E\svu\bait\..\IIS404R_data.mdf' )
Log on    (name='IIS404R_Log' , Filename=' E\svu\bait\....\IIS404R_log.ldf' )

Go
Use IIS404R
Go

Create table ReplicationTable (id int primary key)
Create table noReplicationTable (id int)
```

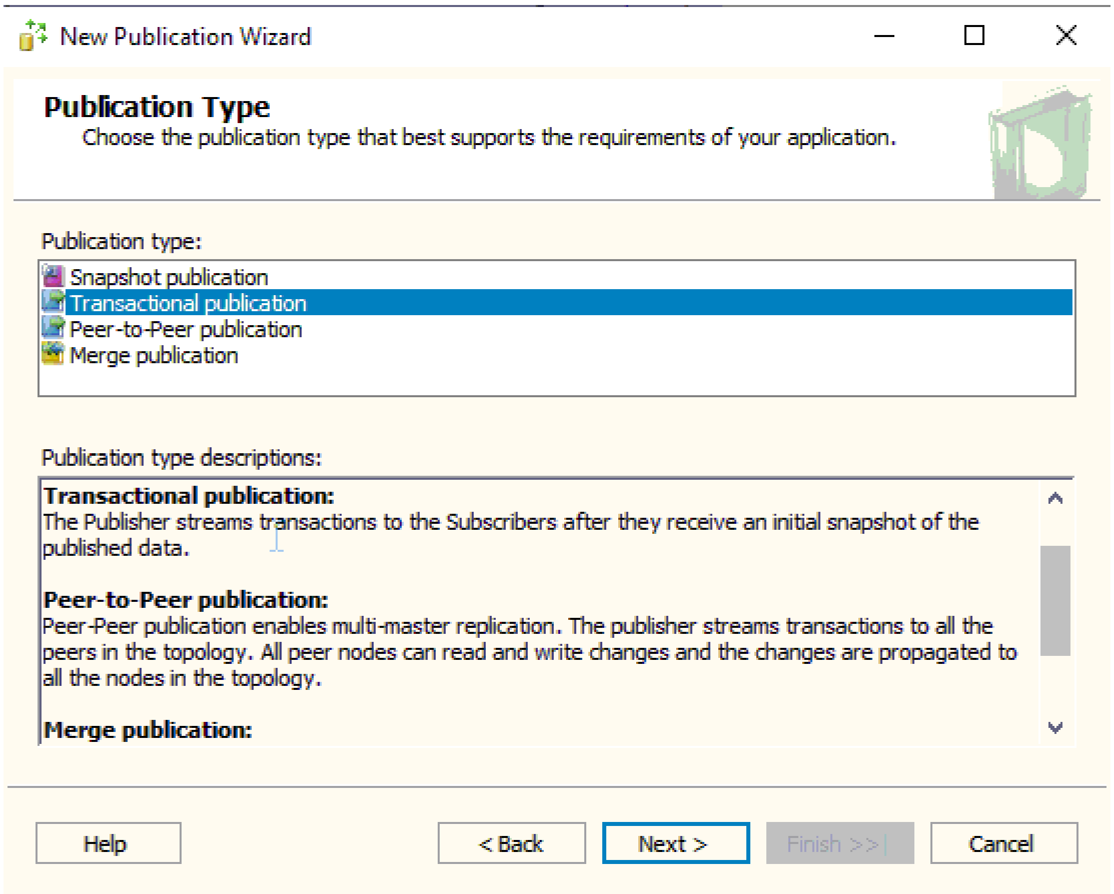
بعد إنشاء جدولين في ال DB, نبدأ بعملية ال Replication

حيث نقوم بإنشاء Publisher, Distributor....

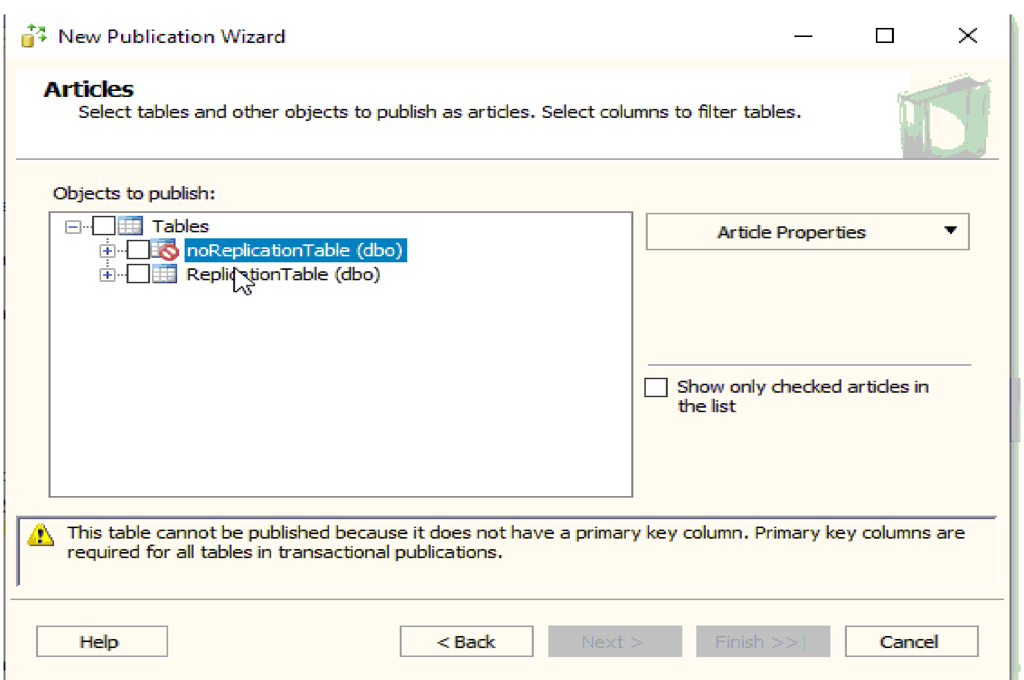
عن طريق مجلد ال Replication , نقوم بإختيار إنشاء New Publisher و من ضمن خطواته يتم إنشاء ال Distributor

و من ضمن الخطوات يتم تحديد موقع حفظ ال Snapshot Replication الأوّلية

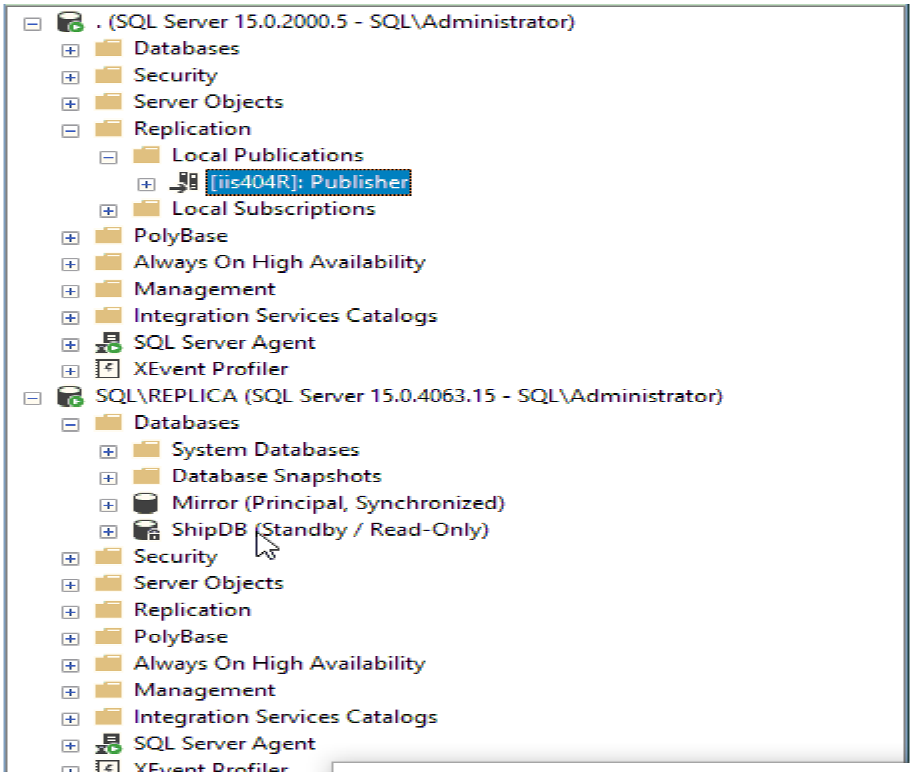
و من ثم تحديد نوع ال Publication : (سنختار نوع ال Transactional):



و من ثم نحدد ال Objects المراد عمل Replication لهم (لن نستطيع تحديد الجدول noReplicationTable بسبب عدم إحتوائه على أي PK)



و عند الإنتهاء سيظهر لدينا ال Publisher في السيرفر الأول كالتالي:



و على السيرفر الثاني نختار من مجلد Replication , خيار New Subscription

حيث نقوم في الخطوات بتحديد ال Publisher المراد الإرتباط معه (نحدد ال Publisher الذي أنشأناه منذ قليل على السيرفر الأول)

ثم نقوم بتحديد ال DB التي سيتم النسخ عليها لدى ال Subscriber (تحديد واحدة موجودة أو إنشاء جديدة)

للتأكد بعد الإنتهاء, نقوم بإدخال قيمة 1000 للجدول ReplicationTable لدى ال Publisher

ثم نستعلم عن معلومات بيانات الجدول نفسه لدى ال Subscriber

Merge replication, like transactional replication, typically starts with a snapshot of the publication database objects and data. Subsequent data changes and schema modifications made at the Publisher and Subscribers are tracked with triggers.

The Subscriber synchronizes with the Publisher when connected to the network and exchanges all rows that have changed between the Publisher and Subscriber since the last time synchronization occurred.

3.3. Database Mirroring

> ⓘ **Note**
>
> This feature will be removed in a future version of Microsoft SQL Server. Avoid using this feature in new development work, and plan to modify applications that currently use this feature. Use Always On availability groups instead.

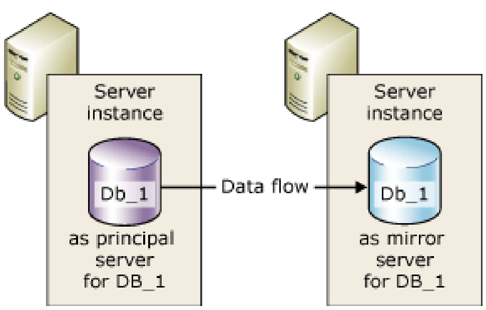
- High-performance mode: Asynchronously and uses only the principal server and mirror server. (With possible data loss) not supported in 2017.
- High-safety mode: Synchronously and, optionally, uses a witness.

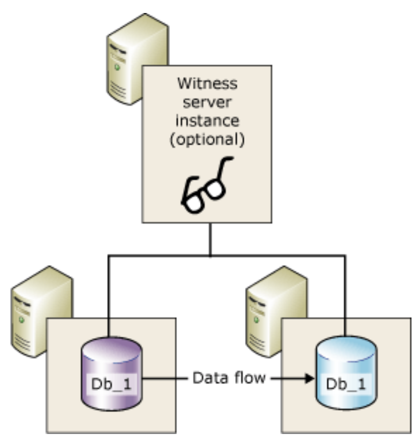
هي صورة عن ال DB, (خاصة صورة عن ال Publisher). و لا تكون Readable, حيث عند وقوع ال Principle Server سيتم عمل ال Mirror Server .

و لها خيار إضافة سيرفر من نوع Witness لكي تصبح عملية ال Fail Over أوتوماتيكية

- لتفعيل ال Mirroring على DB معيّنة, نضغط باليمين عليها و نختار Mirror في النسخ القديمة
و لكن في النسخ الحديثة, يجب القيام بحركة معيّنة عن طريق Script لتفعيلها. (نقوم بإضافة Partner لكل طرف بشكل Script):

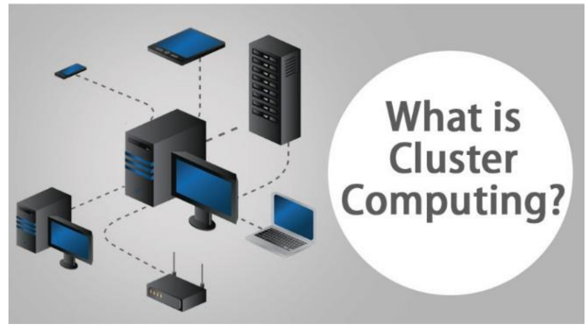
```
Alter Database db_name
Set partner = 'TCP://SQL:5022'
```



WCF & AG

Clustering



Clustering is the connecting of computers (nodes) in a set of two or more nodes, that work together and present themselves to the network as one computer. In most cluster configurations, only one node can be active in a cluster. To ensure that this happens, a quorum instructs the cluster as to which node should be active. It also steps in if there is a communication failure between the nodes.

Each node has a vote in a quorum. However, if there is an even number of nodes, to ensure a simple majority an additional witness must be included in a quorum to allow for a majority vote to take place.

Windows Server Failover Cluster

- ***ال Cluster هو عدة Servers نراهم كواحد فقط***
- ***ال Failover Cluster هي ميّزة في ال Windows Server و موجودة ميّزة مثلها في ال Linux إسمها CRM-Cluster Resource Manager***

As Microsoft describes it:

“Failover clusters provide high availability and scalability to many server workloads. These include server applications such as Microsoft Exchange Server, Hyper-V, Microsoft SQL Server, and file servers. The server applications can run on physical servers or virtual machines. [Windows Server Failover Clustering] can scale to 64 physical nodes and to 8,000 virtual machines.”

(https://technet.microsoft.com/library/hh831579(v=ws.11).aspx).

The terminology here matters. Windows Server Failover Clustering is the name of the technology that underpins a Failover Cluster Instance (FCI), where two or more Windows Server Failover Clustering nodes (computers) are connected together in a Windows Server Failover Clustering resource group and masquerade as a single machine behind a network endpoint called a Virtual Network Name (VNN). A SQL Server service that is installed on an FCI is cluster-aware.

Linux failover clustering with Pacemaker

Instead of relying on Windows Server Failover Clustering, SQL Server on a Linux cluster can make use of any cluster resource manager. Microsoft recommends using Pacemaker because it ships with a number of Linux distributions, including Red Hat and Ubuntu.

3.4. Always On Failover Cluster Instances

-

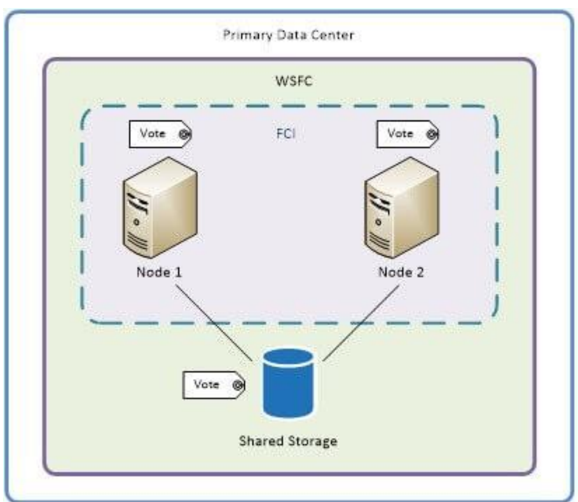
- ***للبدأ بتفعيل ال Always On Failover Cluster نقوم بتنصيب ال SQL Server و نختار خيار ال New SQL failover cluster installation***
على كل عقدة من العقد الخاصة بال Cluster
و بهذه الحالة يتم واحد منهم بالعمل ك Active و الباقي ك Standby
و تكون عملية ال Failover أوتوماتيكية

- ***عملية ال Automatic Failover تعتمد على حجم ال Buffer Pool حيث كل ما زاد حجمه ستزداد الفترة المستخدمة أكثر***
-

You can think of a SQL Server FCI (Failover Cluster Instance) as two or more nodes with shared storage (usually a SAN because it is most likely to be accessed over the network).

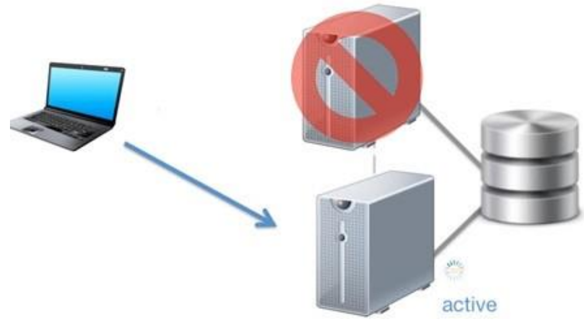
On Windows Server, SQL Server can take advantage of Windows Server Failover Clustering to provide HA (the idea being minimal downtime) at the server-instance level, by creating an FCI of two or more nodes. From the network's perspective (application, end users, and so on), the FCI is presented as a single instance of SQL Server running on a single computer, and all connections point at the VNN.

When the FCI starts, one of the nodes assumes ownership and brings its SQL Server instance online. If a failure occurs on the first node (or there is a planned failover due to maintenance), there are at least a few seconds of downtime, during which the first node cleans up as best it can, and then the second node brings its SQL Server instance online.



Client connections are redirected to the new node after the services are up and running.

When you install SQL Server you select the "New SQL failover cluster installation" option.

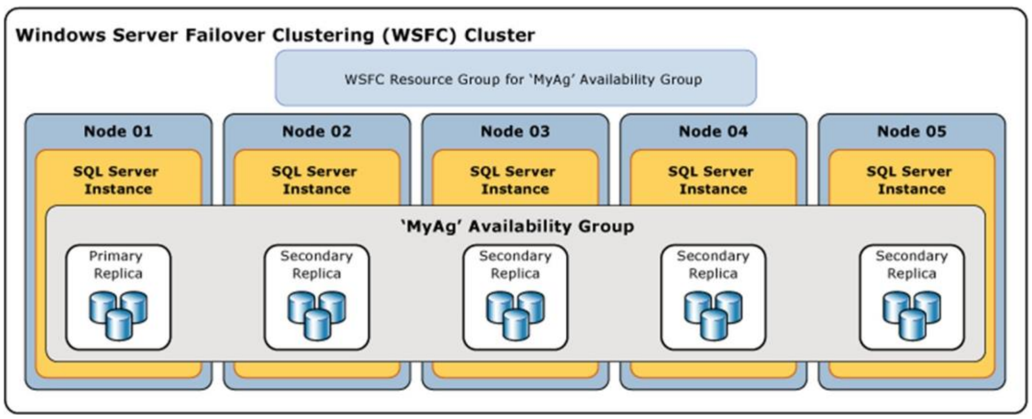

active

How long does the FCI failover take?

During a planned failover, any dirty pages in the buffer pool must be written to the drive; thus, the downtime could be longer than expected on a server with a large buffer pool.

On Linux, the principle is very similar. A cluster resource manager such as Pacemaker manages the cluster, and when a failover occurs, the same process is followed from SQL Server’s perspective, in which the first node is brought down and the second node is brought up to take its place as the owner. The cluster has a virtual IP address, just as on Windows. You must add the virtual network name manually to the DNS server.

3.5. Always On availability groups AG



- تكون هذه التقنية على مستوى ال Server و ال Database
- يكون لدينا Primary Replica عليها ال DB التي تكون قابلة للقراءة و الكتابة , بالإضافة لحد 8 Replicas إضافيّة (ممكن أن يكونو Read-Only و ممكن أن يكونو حتى غير قابلين للقراءة, كأن نستخدمهم لإستخدامات معيّنة)

حيث حين وضع ال Secondary Replica ك Read-Only يتم توزيع الحمل بين ال Replicas عند طلب قراءة من ال DB (Load Balancing)

و يكون ال Failover أيضا اوتوماتيكي

- لها نوعان:

1) Asynchronous (غير متزامن) يستخدم مع المسافات البعيدة و يعتبر من حلول ال DR
2) Synchronous (متزامن) يستخدم مع المسافات القريبة و يعتبر من حلول ال HA

A high-availability and disaster-recovery solution that provides an enterprise-level alternative to database mirroring. It maximizes the availability of a set of user databases for an enterprise.

Supports a failover environment for a discrete set of user databases, known as availability databases that fail over together. Supports a set of read-write primary databases and one to eight sets of corresponding secondary databases. Optionally, secondary databases can be made available for read-only access and/or some backup operations. An availability group fails over at the level of an availability replica. Installing SQL Server is by using install standalone instance per replica.

3.5.1. AG Terms & Definitions

- Availability group: A container for a set of databases, availability databases, that fail over together.
- Availability database: A database that belongs to an availability group
- Primary database: The read-write copy of an availability database.

- Secondary database: A read-only copy of an availability database.
- Availability replica: An instantiation of an availability group that is hosted by a specific instance of SQL Server and maintains a local copy of each availability database that belongs to the availability group.
- Two types of availability replicas exist: a single primary replica and one to eight secondary replicas.
- Primary replica: The availability replica that makes the primary databases available for read-write connections from clients and, also, sends transaction log records for each primary database to every secondary replica.
- Secondary replica: An availability replica that maintains a secondary copy of each availability database, and serves as a potential failover targets for the availability group. Optionally, a secondary replica can support read-only access to secondary databases can support creating backups on secondary databases.
- Availability group listener: A server name to which clients can connect in order to access a database in a primary or secondary replica of an Always On availability group. Availability group listeners direct incoming connections to the primary replica or to a read-only secondary replica.
- **AG** Supports alternative availability modes, as follows:
    - Asynchronous-commit mode: a disaster-recovery solution that works well when the availability replicas are distributed over considerable distances.
    - Synchronous-commit mode: emphasizes high availability and data protection over performance, at the cost of increased transaction latency.

3.5.2. AG Failover Types

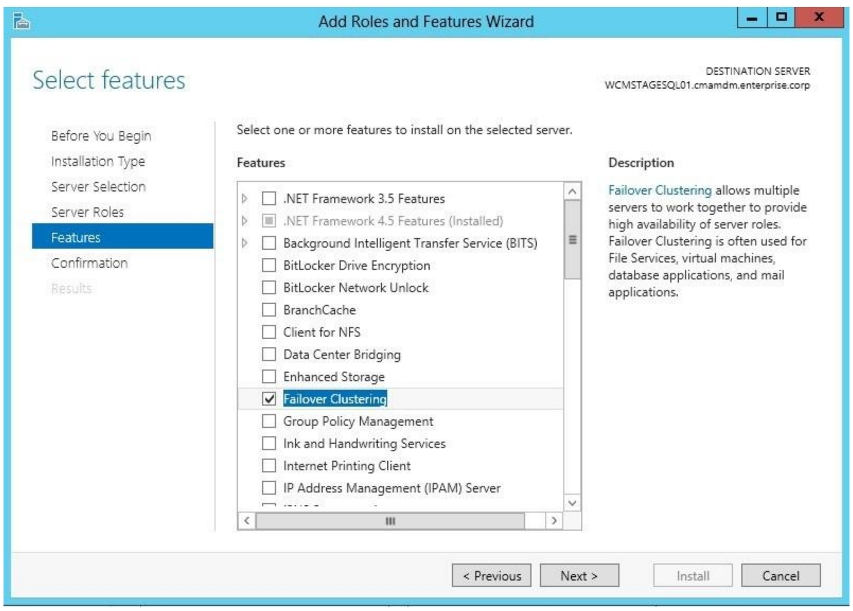
Planned manual failover (without data loss): Occurs after a database administrator issues a failover command and causes a synchronized secondary replica to transition to the primary role (with guaranteed data protection) and the primary replica to transition to the secondary role. A manual failover requires that both the primary replica and the target secondary replica are running under synchronous-commit mode, and the secondary replica must already be synchronized.

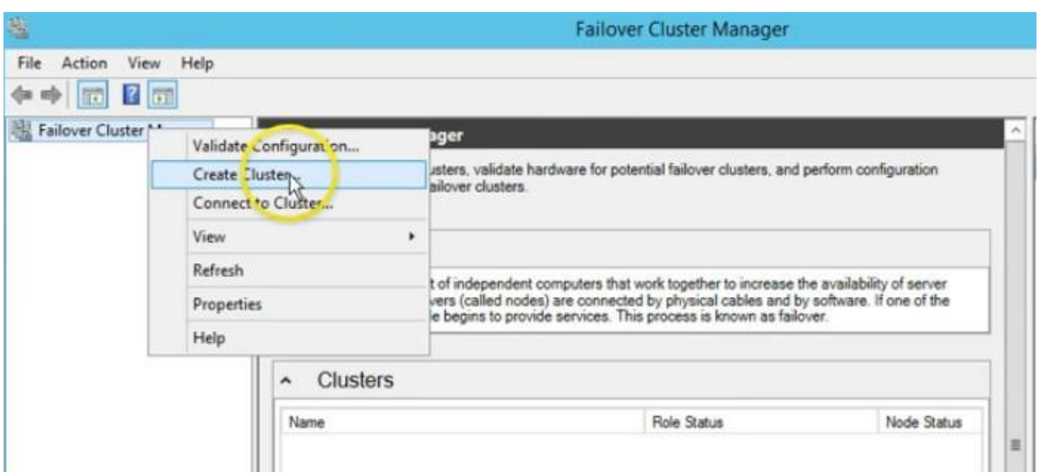
Automatic failover (without data loss): Occurs in response to a failure that causes a synchronized secondary replica to transition to the primary role (with guaranteed data protection). When the former primary replica becomes available, it transitions to the secondary role. Automatic failover requires that both the primary replica and the target secondary replica are running under synchronous-commit mode with the failover mode set to "Automatic". In addition, the secondary replica must already be synchronized, have WSFC quorum, and meet the conditions specified by the flexible failover policy of the availability group.

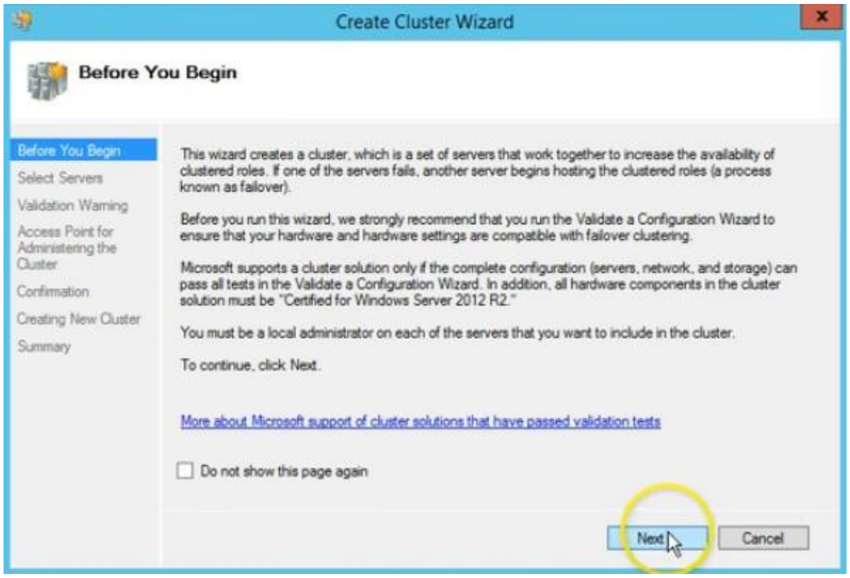
3.5.3. Configuring AlwaysON Demo

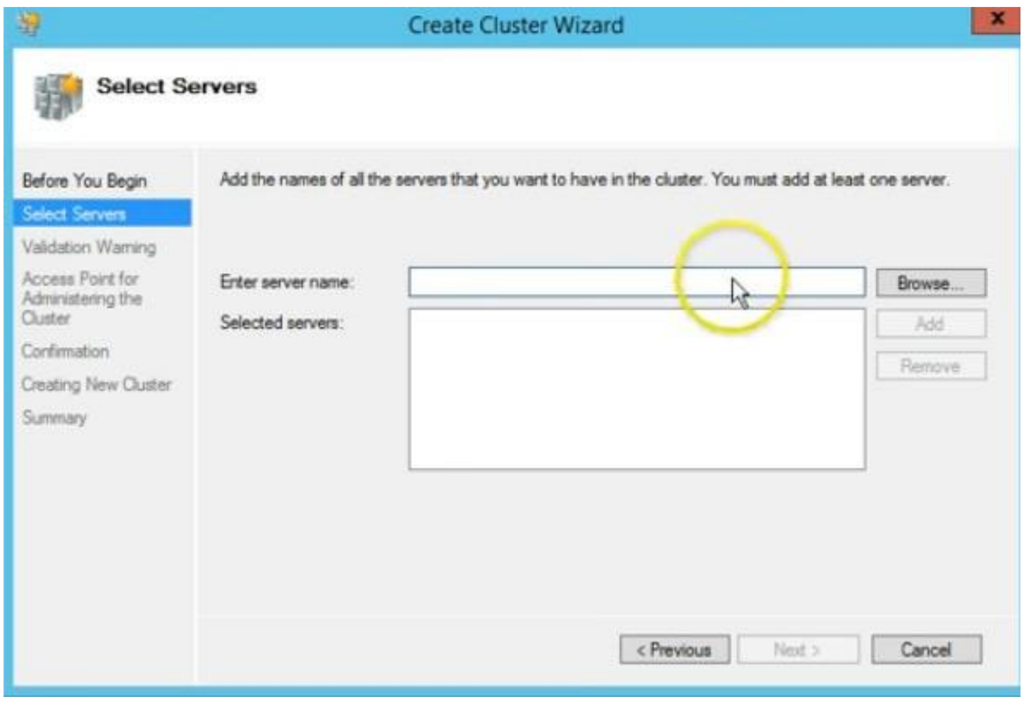
مثال عن تطبيق ال Failover Cluster:

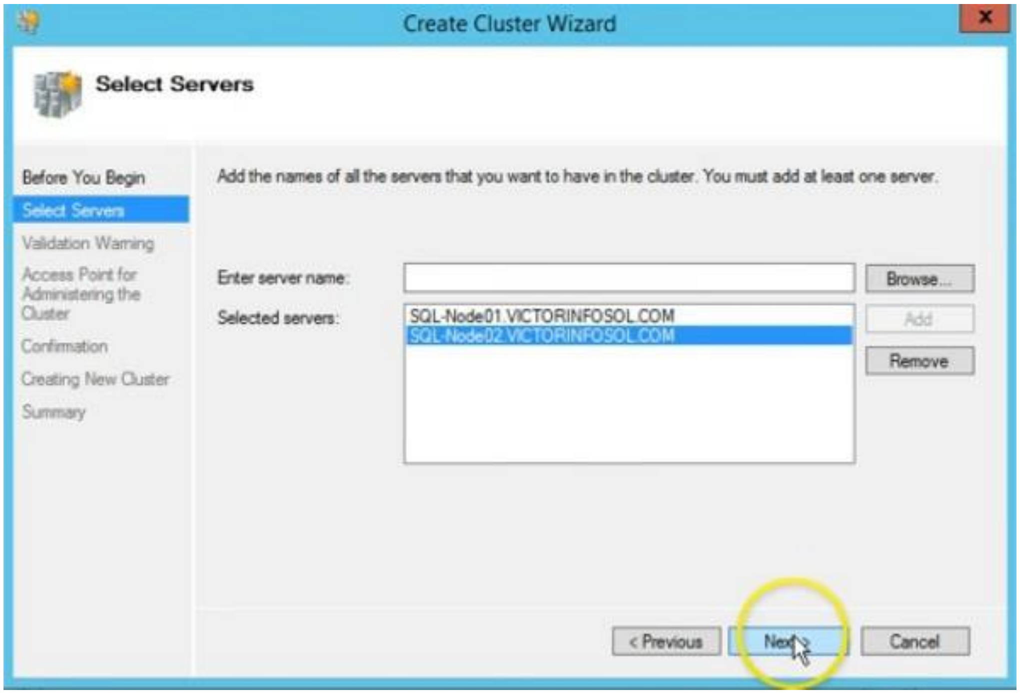
1. We need 3 Servers running Windows Server.
2. One is a DC, the others are SQL Nodes (SQLNode01, SQLNode02).
3. Install and Configure Windows failover cluster on the nodes.



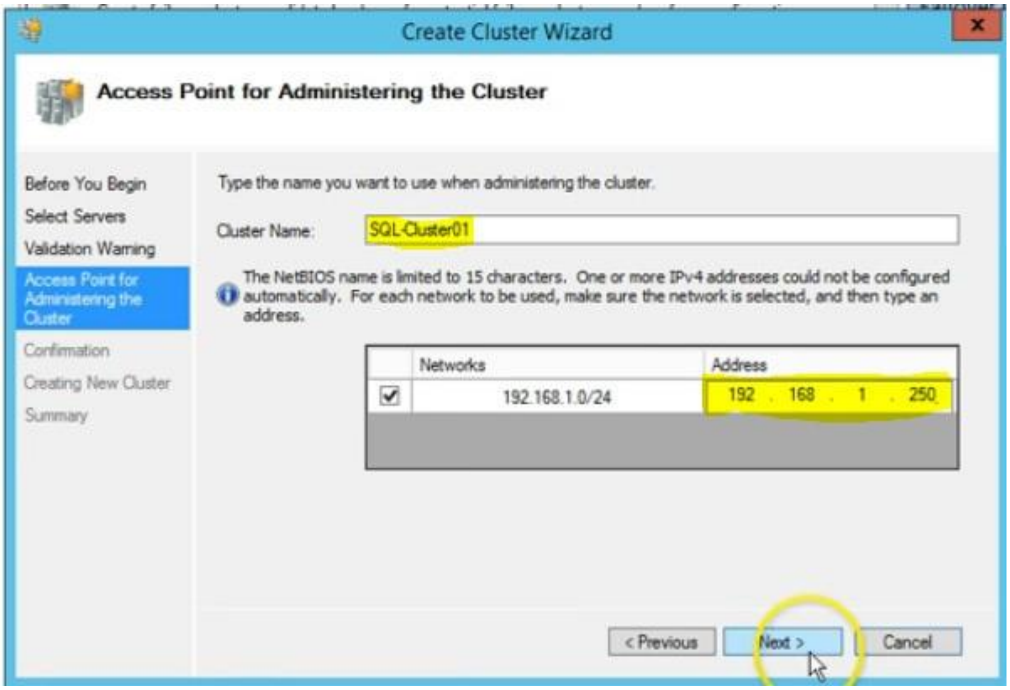


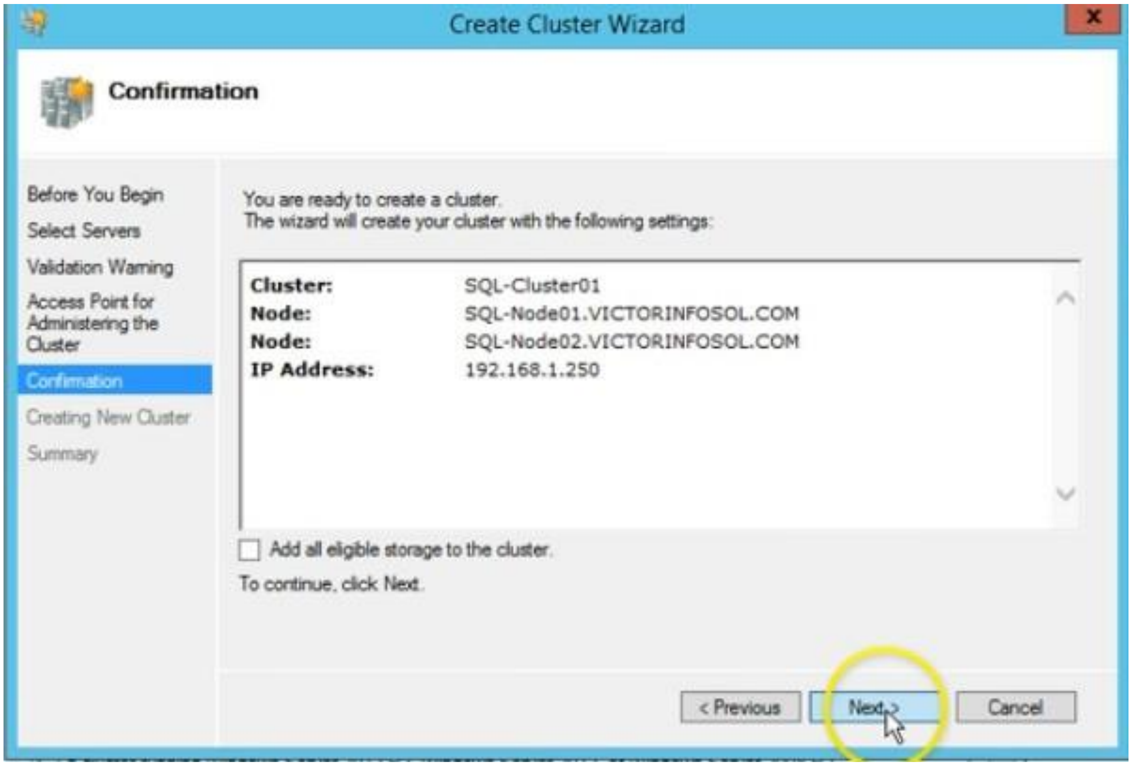

Create Cluster Wizard
Before You Begin
Before You Begin
Select Servers
Validation Warning
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
This wizard creates a cluster, which is a set of servers that work together to increase the availability of clustered roles. If one of the servers fails, another server begins hosting the clustered roles (a process known as failover).
Before you run this wizard, we strongly recommend that you run the Validate a Configuration Wizard to ensure that your hardware and hardware settings are compatible with failover clustering.
Microsoft supports a cluster solution only if the complete configuration (servers, network, and storage) can pass all tests in the Validate a Configuration Wizard. In addition, all hardware components in the cluster solution must be "Certified for Windows Server 2012 R2."
You must be a local administrator on each of the servers that you want to include in the cluster.
To continue, click Next.
More about Microsoft support of cluster solutions that have passed validation tests
Do not show this page again
Next
Cancel

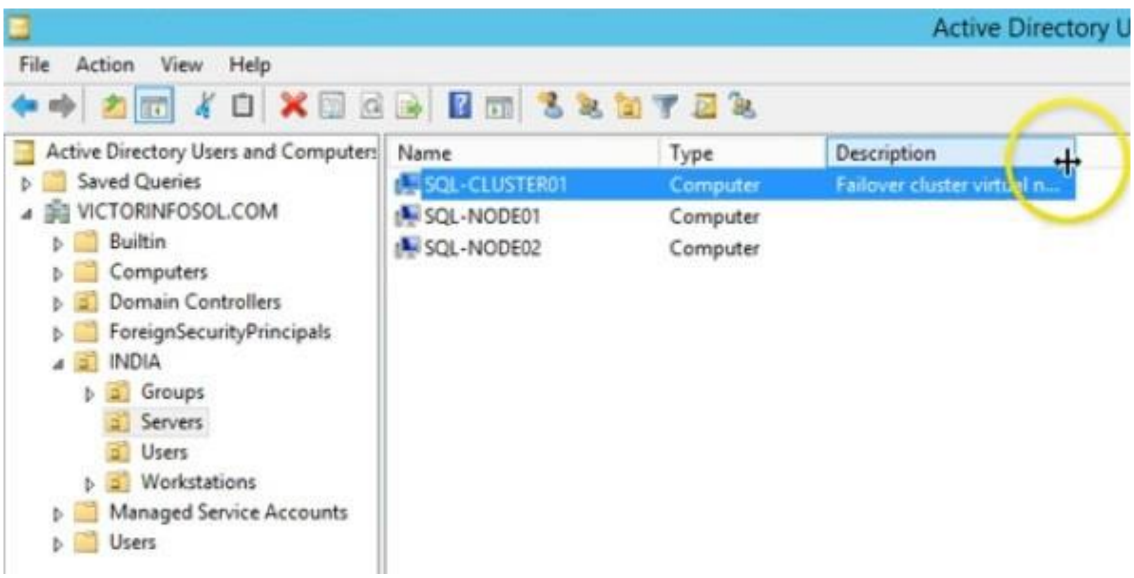

Create Cluster Wizard
Select Servers
Before You Begin
Select Servers
Validation Warning
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
Add the names of all the servers that you want to have in the cluster. You must add at least one server.
Enter server name:
Selected servers:
Browse...
Add
Remove
< Previous
Next >
Cancel

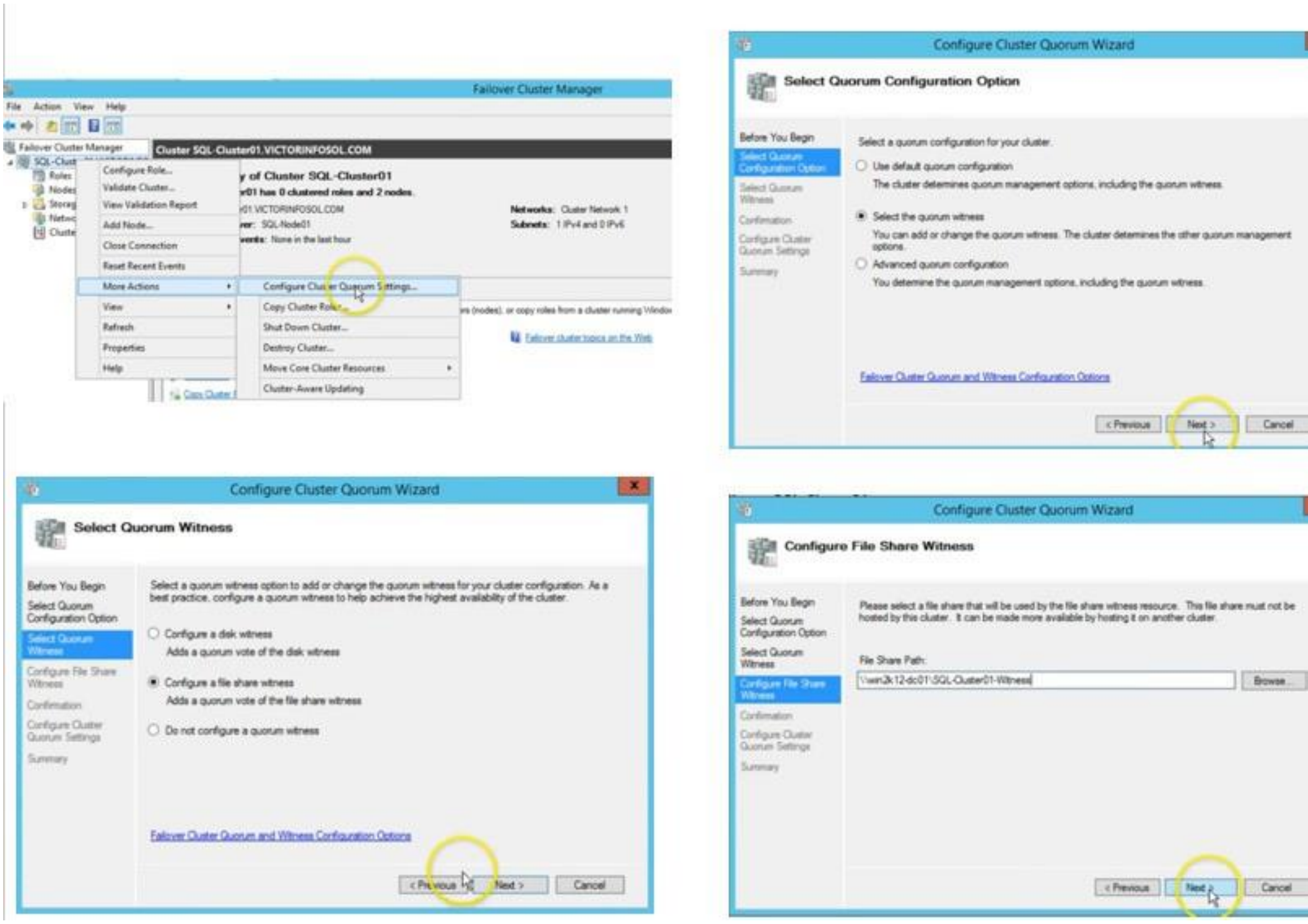

Create Cluster Wizard
Select Servers
Before You Begin
Select Servers
Validation Warning
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
Add the names of all the servers that you want to have in the cluster. You must add at least one server.
Enter server name:
Browse...
Selected servers:
SQL-Node01.VICTORINFOSOL.COM
SQL-Node02.VICTORINFOSOL.COM
Add
Remove
< Previous
Cancel


Create Cluster Wizard
Validation Warning
Before You Begin
Select Servers
Validation Warning
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
For the servers you selected for this cluster, the reports from cluster configuration validation tests appear to be missing or incomplete. Microsoft supports a cluster solution only if the complete configuration (servers, network and storage) can pass all the tests in the Validate a Configuration wizard.
Do you want to run configuration validation tests before continuing?
Yes. When I click Next, run configuration validation tests, and then return to the process of creating the cluster.
No. I do not require support from Microsoft for this cluster, and therefore do not want to run the validation tests. When I click Next, continue creating the cluster.
More about cluster validation tests
< Previous
Next
Cancel

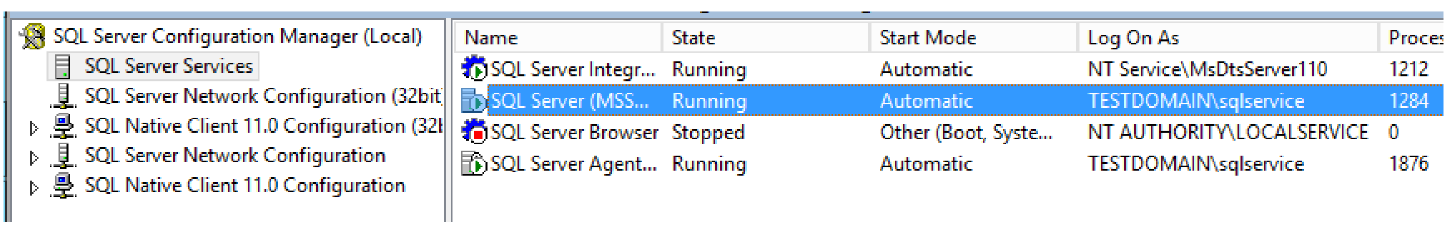

Create Cluster Wizard
Access Point for Administering the Cluster
Before You Begin
Select Servers
Validation Warning
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
Type the name you want to use when administering the cluster.
Cluster Name:
SQL-Cluster01
The NetBIOS name is limited to 15 characters. One or more IPv4 addresses could not be configured automatically. For each network to be used, make sure the network is selected, and then type an address.
Networks
Address
192.168.1.0/24
192 . 168 . 1 . 250
< Previous
Next >
Cancel


Create Cluster Wizard
Confirmation
Before You Begin
Select Servers
Validation Warning
Access Point for Administering the Cluster
Confirmation
Creating New Cluster
Summary
You are ready to create a cluster.
The wizard will create your cluster with the following settings:
Cluster: SQL-Cluster01
Node: SQL-Node01.VICTORINFOSOL.COM
Node: SQL-Node02.VICTORINFOSOL.COM
IP Address: 192.168.1.250
Add all eligible storage to the cluster.
To continue, click Next.
< Previous
Next >
Cancel

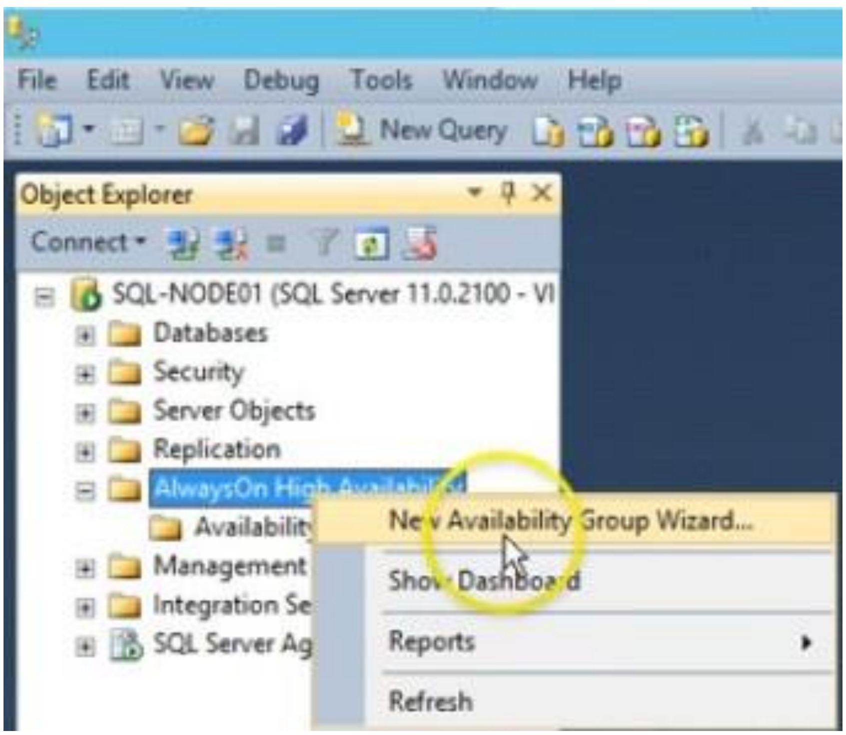

Active Directory U
File Action View Help
Active Directory Users and Computers
Saved Queries
VICTORINFOSOL.COM
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipals
INDIA
Groups
Servers
Users
Workstations
Managed Service Accounts
Users
Name
Type
Description
SQL-CLUSTER01
Computer
Failover cluster virtual n...
SQL-NODE01
Computer
SQL-NODE02
Computer

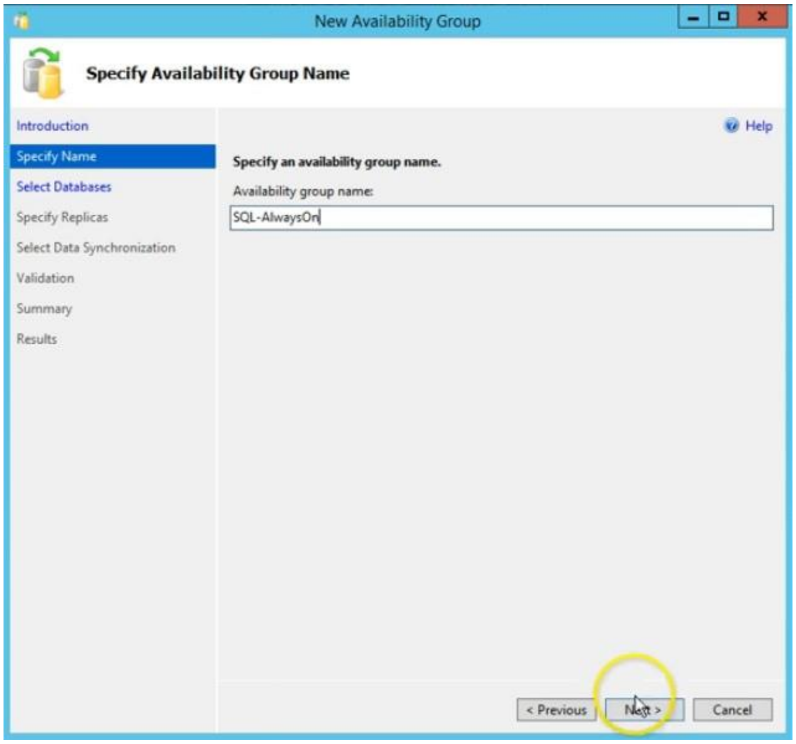


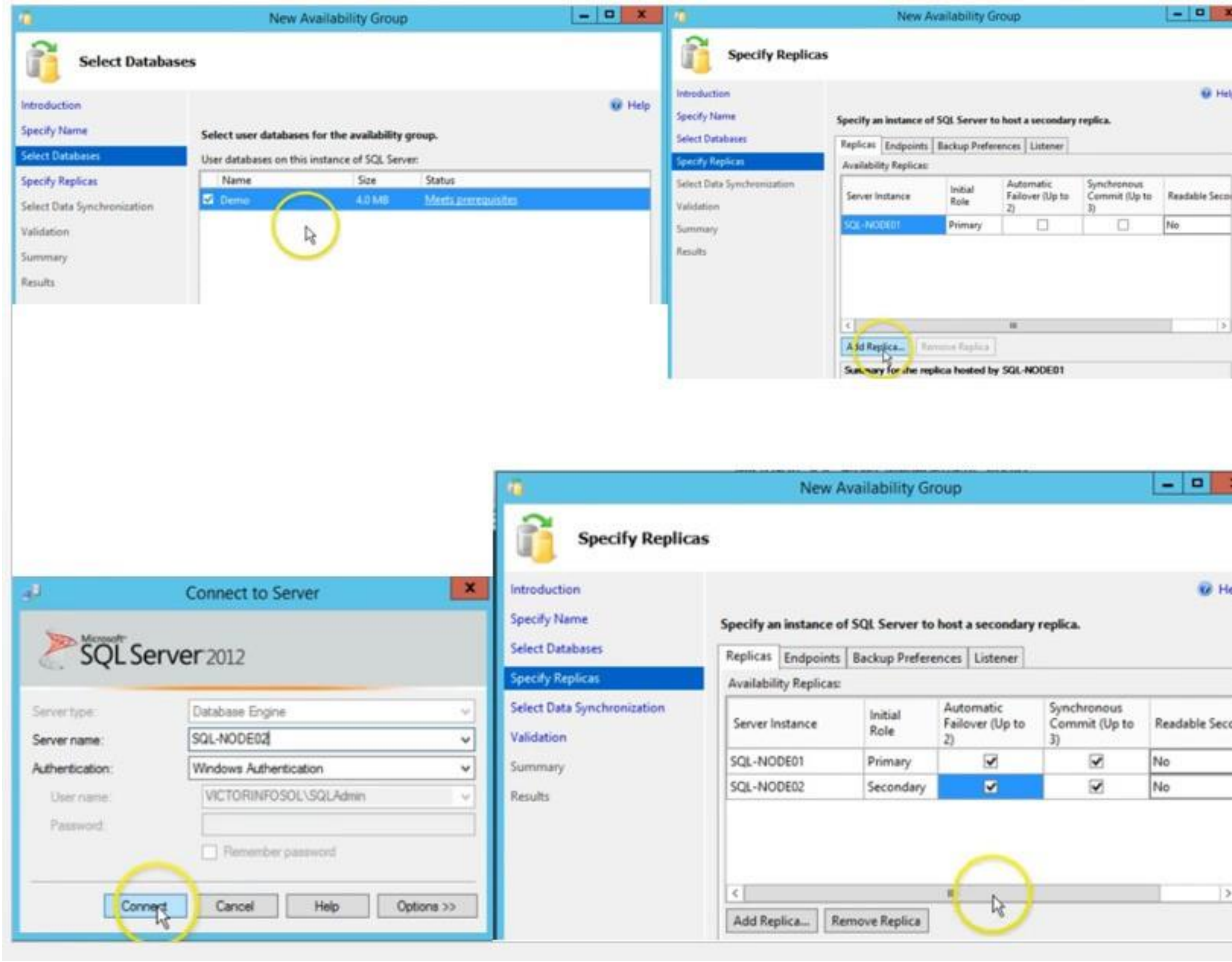
4. If Firewall is active then add port #1433 & #5022 to the firewall.
5. Install SQL Server on both nodes.
6. Create a test Database on SQLNode01 then take a full backup and restore it on SQLNode02
7. From SQL Server Configuration manager in both nodes, Set the properties of service and enable AlwaysON.

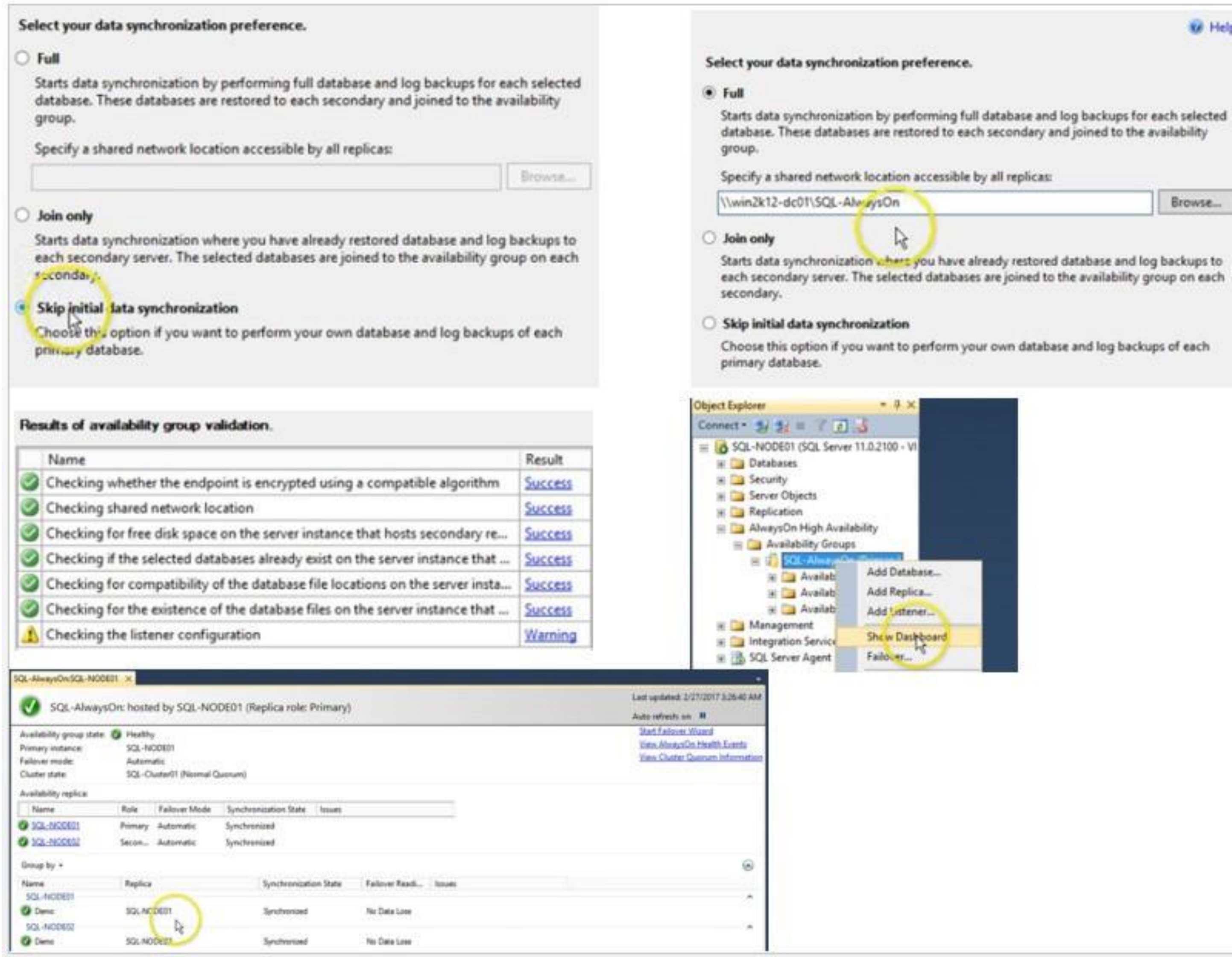


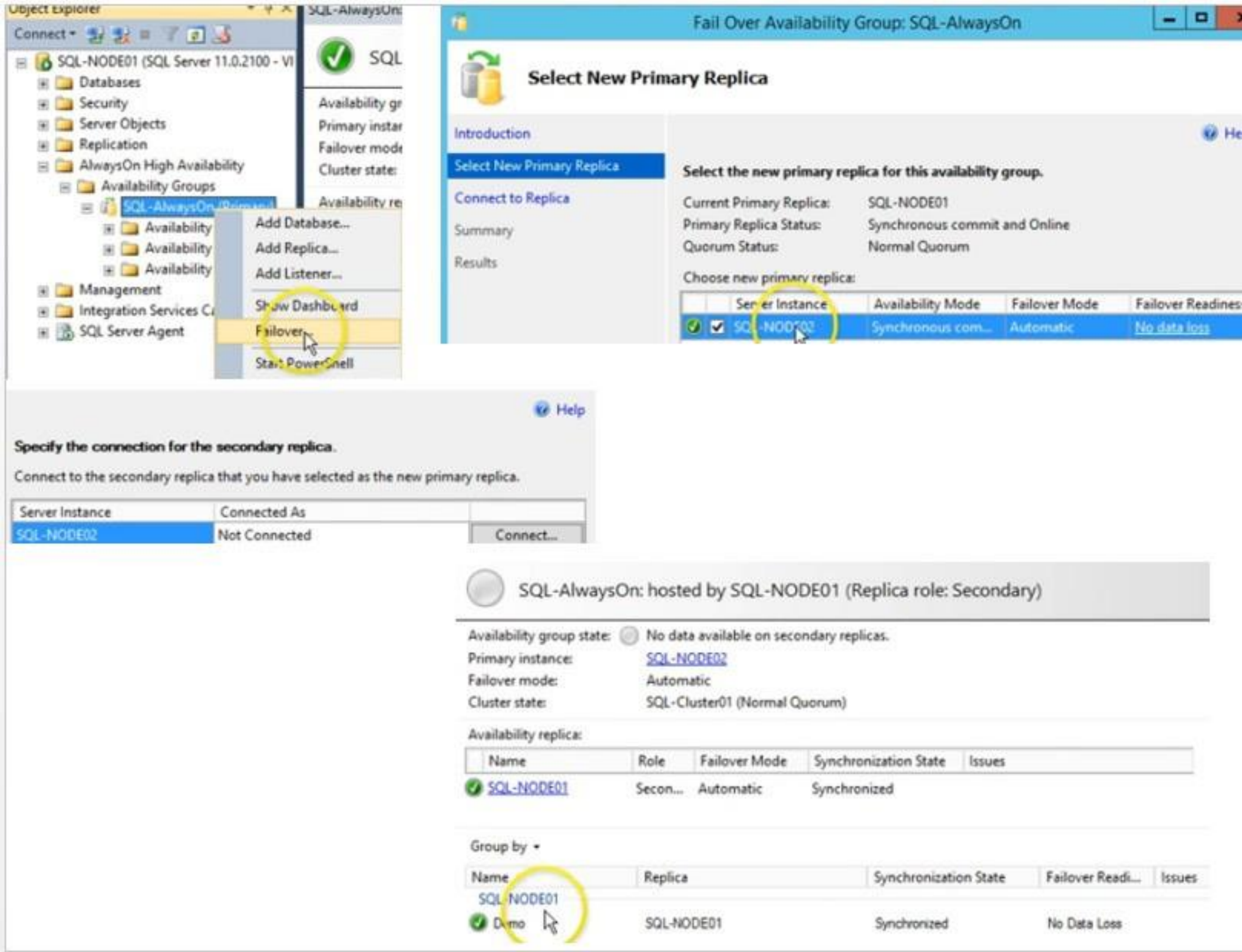
❖ Create AlwaysON availability Group on SQLNode01




New Availability Group
Specify Availability Group Name
Introduction
Specify Name
Select Databases
Specify Replicas
Select Data Synchronization
Validation
Summary
Results
Help
Specify an availability group name.
Availability group name:
SQL-AlwaysOn
< Previous
Next >
Cancel


New Availability Group
Select Databases
Introduction
Specify Name
Select Databases
Specify Replicas
Select Data Synchronization
Validation
Summary
Results
Help
Select user databases for the availability group.
User databases on this instance of SQL Server:
Name
Size
Status
Demo
4.0 MB
Meets prerequisites
New Availability Group
Specify Replicas
Specify an instance of SQL Server to host a secondary replica.
Replicas
Endpoints
Backup Preferences
Listener
Availability Replicas:
Server Instance
Initial Role
Automatic Failover (Up to 2)
Synchronous Commit (Up to 3)
SQL-NODE01
Primary
No
Add Replica...
Remove Replica
Summary for the replica hosted by SQL-NODE01
Connect to Server
Microsoft SQL Server 2012
Server type:
Database Engine
Server name:
SQL-NODE02
Authentication:
Windows Authentication
User name:
VICTORINFOSOL\SQLAdmin
Password:
Remember password
Connect
Cancel
Help
Options >>
SQL-NODE02
Secondary


Select your data synchronization preference.
Full
Starts data synchronization by performing full database and log backups for each selected database. These databases are restored to each secondary and joined to the availability group.
Specify a shared network location accessible by all replicas:
Browse...
Join only
Starts data synchronization where you have already restored database and log backups to each secondary server. The selected databases are joined to the availability group on each secondary.
Skip initial data synchronization
Choose this option if you want to perform your own database and log backups of each primary database.
Select your data synchronization preference.
Full
Starts data synchronization by performing full database and log backups for each selected database. These databases are restored to each secondary and joined to the availability group.
Specify a shared network location accessible by all replicas:
\\win2k12-dc01\SQL-AlwaysOn
Browse...
Join only
Starts data synchronization where you have already restored database and log backups to each secondary server. The selected databases are joined to the availability group on each secondary.
Skip initial data synchronization
Choose this option if you want to perform your own database and log backups of each primary database.
Results of availability group validation.
Name
Result
Checking whether the endpoint is encrypted using a compatible algorithm
Success
Checking shared network location
Success
Checking for free disk space on the server instance that hosts secondary re...
Success
Checking if the selected databases already exist on the server instance that ...
Success
Checking for compatibility of the database file locations on the server insta...
Success
Checking for the existence of the database files on the server instance that ...
Success
Checking the listener configuration
Warning
Object Explorer
Connect
SQL-NODE01 (SQL Server 11.0.2100
Databases
Security
Server Objects
Replication
AlwaysOn High Availability
Availability Groups
Management
Integration Services
SQL Server Agent
Add Database...
Add Replica...
Add Listener...
Show Dashboard
Failover...
SQL-AlwaysOn: hosted by SQL-NODE01 (Replica role: Primary)



References

LeBlanc. P, Assaf. W, Jackson. B, Curnutt M; "**SQL Server 2017 Administration Inside Out**", Microsoft Press (2018).

https://docs.microsoft.com/en-us/sql/sql-server/